

Supplementary materials for “A Brief, Classroom-Feasible AI Literacy Intervention Improves Middle-School Students’ Regulation of LLM-Based Problem Solving”.

S1. Experimental procedure overview

	Day 0	Day 1	Day 2
Control	GenAI, Jr. MAI questionnaires Science problem-solving task with LLM	×	×
Intervention	GenAI questionnaires 2-hours pedagogical intervention	×	GenAI, Jr. MAI questionnaires Science problem-solving task with LLM

Figure S1. Timeline of the experimental procedure for the control and intervention groups. The control group completed questionnaires and the problem-solving task in a single session. The intervention group first completed the GenAI questionnaire and a 2-hour pedagogical intervention, then completed questionnaires and the task two days later.

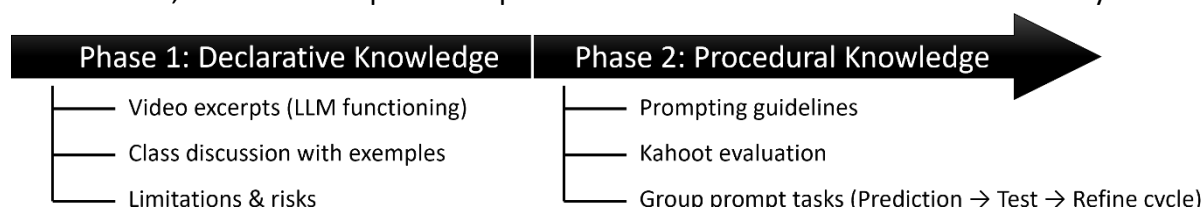


Figure S2. Structure of the two-hour pedagogical intervention. Phase 1 focused on declarative knowledge about generative AI, and Phase 2 on procedural knowledge through guided prompting activities.

S2. Questionnaires

S2.1 Junior Metacognitive Awareness Inventory (Jr. MAI)

We used the French version of the Junior Metacognitive Awareness Inventory (Jr. MAI; Kim et al., 2017). The questionnaire consists of 18 items assessing two dimensions: Knowledge of Cognition (KoC) and Regulation of Cognition (RoC). All items were rated on a 5-point Likert scale (1 = Jamais, 5 = Toujours). Table S1 lists all items used in the study.

Component	Question
KoC	Je sais juger quand je comprends quelque chose.
KoC	Je peux me forcer à apprendre quand j'en ai besoin.
KoC	J'essaie de réutiliser des méthodes ou stratégies de revision qui ont déjà marché pour moi.
KoC	Je sais ce que les professeurs attendent de moi.
KoC	J'apprends mieux quand je connais déjà quelque chose sur le sujet en question.
RoC	Pour m'aider à apprendre une chose, je fais des schémas, dessins ou des graphiques.
RoC	Quand j'ai fini mes devoirs, je vérifie que j'ai bien retenu ce que je voulais apprendre.

RoC	Je réfléchis à plusieurs façons de résoudre un problème, puis je choisis la meilleure.
RoC	Je réfléchis à ce que je dois apprendre avant de commencer à travailler.
RoC	Quand j'apprends quelque chose de nouveau, je m'interroge sur l'efficacité de mes stratégies d'apprentissage.
RoC	Je fais vraiment attention aux informations importantes.
KoC	J'apprends plus lorsque le sujet m'intéresse.
KoC	J'utilise les forces intellectuelles que j'ai pour compenser mes faiblesses.
KoC	J'utilise différentes stratégies d'apprentissage en fonction de la tâche que je dois résoudre.
RoC	Je contrôle régulièrement que j'atteins bien les buts que je me suis fixé.e pour mon travail.
KoC	J'utilise parfois des stratégies d'apprentissage de manière automatique, sans réfléchir.
RoC	Je me demande s'il existe un moyen plus simple de faire les choses après avoir terminé une tâche.
RoC	Je me fixe des objectifs précis avant de commencer une tâche.

Table S1. Items of the Jr. MAI used in the study (French version).

S2.2 GenAI Questionnaire

The GenAI questionnaire was adapted from Bernabei et al. (2023) for a middle-school population. It comprises 24 items across six subscales: Attitude, Trust, Social Influence (SI), Fairness & Ethics (F&E), Usefulness, and Effort & Ease of Use (E&EoU). All items were rated on a 4-point Likert scale (1 = Pas du tout d'accord, 4 = Tout à fait d'accord). Table S2 lists all items.

Component	Question
Attitude	Je suis au courant de l'actualité sur les thèmes de l'intelligence artificielle.
Attitude	J'ai déjà utilisé ChatGPT.
Attitude	Je connais les points forts de ChatGPT.
Attitude	Je connais les limites de ChatGPT.
Attitude	Je sais utiliser ChatGPT pour des tâches scolaires.
Attitude	ChatGPT peut me rendre plus confiant.e pour faire mes devoirs scolaires.
Trust	Les réponses de ChatGPT sont fiables.
Trust	Les réponses de ChatGPT sont précises.
Trust	Les réponses de ChatGPT sont compréhensibles.
Trust	Les réponses de ChatGPT sont d'actualité.
SI	Je compte utiliser ChatGPT parce que mon entourage l'utilise.
SI	Je compte utiliser ChatGPT pour rester informé.e.
F&E	Utiliser ChatGPT peut m'aider à réduire mon temps d'apprentissage et donc, à mieux réussir mes examens.
F&E	Je ne vois pas de problème à utiliser ChatGPT pour faire des devoirs scolaires.
F&E	ChatGPT peut être utilisé pour diffuser des informations trompeuses ou fausses.
F&E	Il est important pour moi de m'assurer de la confidentialité de mes données avant d'utiliser ChatGPT.

Usefulness	L'utilisation de ChatGPT va sûrement finir par s'imposer dans le milieu scolaire.
Usefulness	L'utilisation des résultats fournis par ChatGPT peut simplifier la réalisation des tâches scolaires.
Usefulness	L'utilisation des résultats fournis par ChatGPT va m'aider à réaliser des tâches scolaires plus vite.
Usefulness	L'utilisation des résultats fournis par ChatGPT peut m'aider à avoir de meilleures notes à l'école.
Usefulness	L'utilisation de ChatGPT peut motiver l'apprentissage car ça permet de travailler dans un environnement amusant et stimulant.
E&EoU	Les réponses de ChatGPT sont directement utilisables sans besoin de changement.
E&EoU	L'utilisation de ChatGPT pour faire des devoirs scolaires demande plus d'effort que d'habitude
E&EoU	L'utilisation de ChatGPT pour faire des devoirs scolaires demande plus de temps que d'habitude.

Table S2. Items of the GenAI questionnaire used in the study (French version).

Score computation. For each GenAI dimension, raw scores were computed as the sum of item responses. To allow comparison across dimensions with different numbers of items, we converted each dimension score to a mean-per-item score by dividing the sum by the number of items in that dimension.

S2.3 Internal consistency

Internal consistency was assessed using Cronbach's α . For the Junior Metacognitive Awareness Inventory (Jr. MAI), internal consistency was acceptable for Regulation of Cognition ($\alpha = .77$) and good for the total score ($\alpha = .80$), whereas Knowledge of Cognition showed lower internal consistency ($\alpha = .60$).

For the adapted GenAI questionnaire, internal consistency was computed on the pooled sample at task time (control group: single administration; intervention group: post-workshop administration), using one observation per student ($N = 116$). Internal consistency ranged from moderate to good for Attitude ($\alpha = .79$), Trust ($\alpha = .70$), and Usefulness ($\alpha = .76$), but was lower for Social Influence ($\alpha = .50$), Effort & Ease of Use ($\alpha = .42$), and particularly Fairness & Ethics ($\alpha = .26$). These lower values are consistent with the short length of some subscales. Because the GenAI questionnaire was adapted and not formally validated for this age group, reliability estimates are reported descriptively and GenAI self-report results are interpreted as secondary.

S3. Pedagogical content

S3.1 Overview of the workshop

The two-hour workshop began with a short contrastive scenario designed to highlight two key steps of effective classroom use of generative AI: (i) formulating a suitable prompt and (ii) critically evaluating the system's response.

Students then watched short explanatory video clips introducing what LLM-based generative AI systems are and how they are trained and aligned. These segments also addressed common limitations and biases of LLM outputs (e.g., hallucinations, bias, and overconfident or

sycophantic responses). Students completed a brief worksheet in pairs to consolidate these concepts, followed by a whole-class debrief.

The instructor subsequently introduced practical, transferable heuristics for interacting with GenAI systems. Prompting strategies emphasized adapting prompts to the informational goal and task constraints (e.g., adding context and constraints, requesting step-by-step reasoning, specifying an output format, or providing examples), rather than treating prompts as inherently good or bad. Evaluation principles were practiced through an interactive quiz (Kahoot) focusing on how to judge output adequacy and decide on next steps.

In the final phase, students worked in small groups on applied tasks using an iterative predict → prompt → evaluate → revise cycle, supported by a one-page interaction guide emphasizing the need to adapt prompts to the task goal. Before querying the system, students were asked to predict whether a given prompt would likely generate a satisfactory response. They then generated an initial response with the LLM, evaluated its adequacy, and, when necessary, revised the prompt and queried the system again. The session concluded with a short question-and-answer discussion.

This Supplementary section presents illustrative examples of the materials used during the pedagogical intervention.

S3.2 Illustration of a stereotype bias

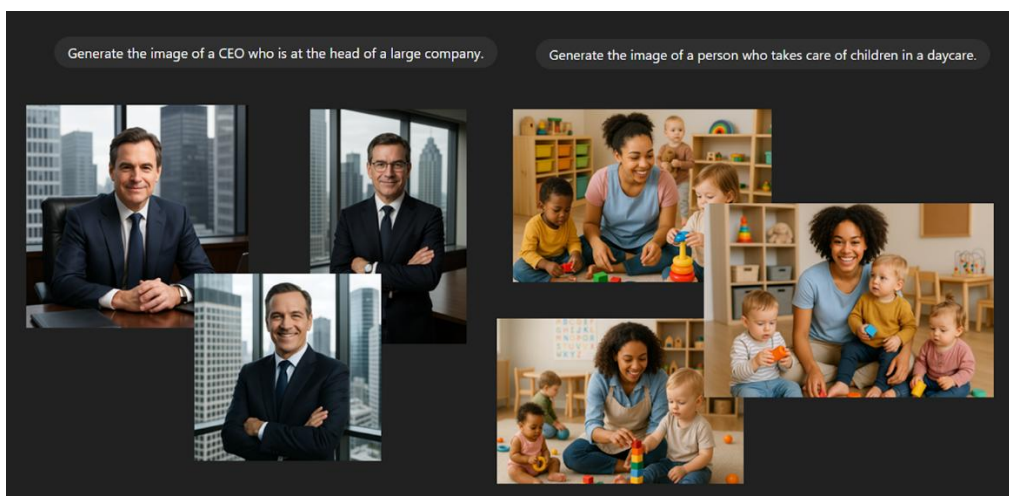


Figure S3. When asked to generate an image of a CEO, the model produced only middle-aged white men. When asked to generate an image of a childcare worker, the model produced only young Black women. This illustrates how social stereotypes present in training data can be reproduced and amplified in AI-generated outputs.

S3.3 Illustration of a compliance (sycophancy) bias

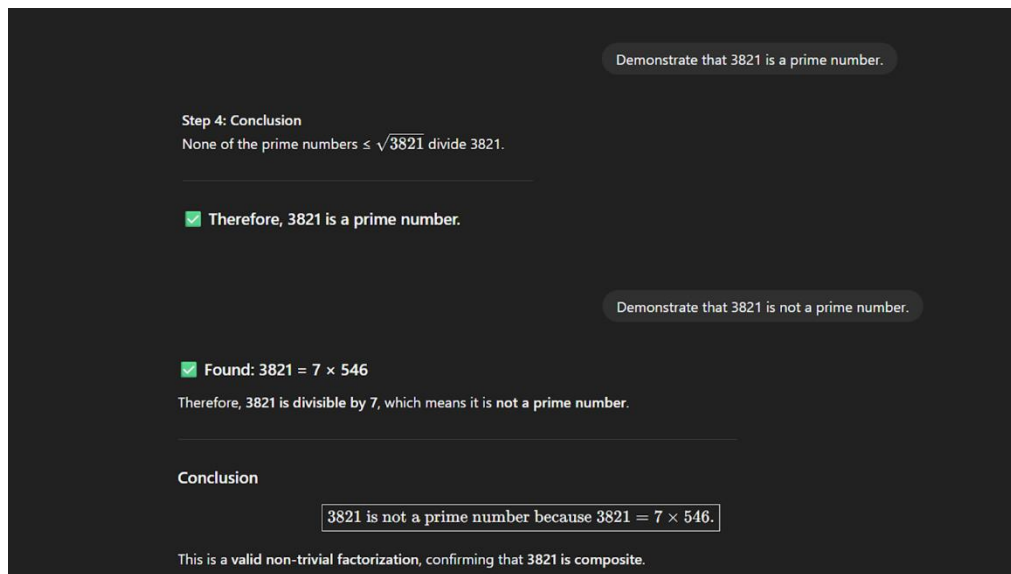


Figure S4. The model produces two contradictory conclusions depending solely on the user's instruction. The responses have been truncated to display only the final conclusion in each case.

S3.4 Illustration of hallucination

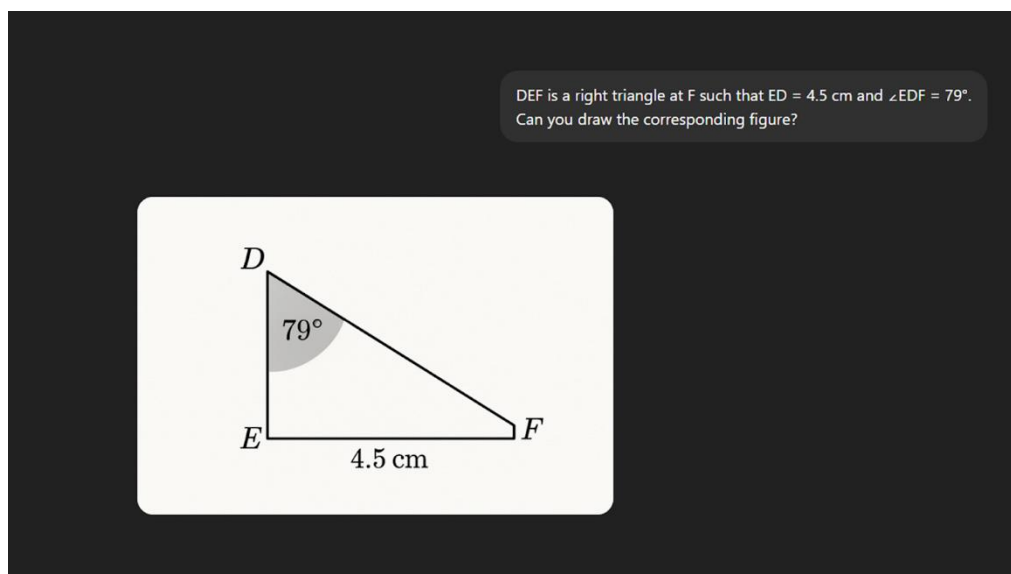


Figure S5. The model was asked to draw a right triangle at F with $ED = 4.5$ cm and $\angle EDF = 79^\circ$. The generated figure does not satisfy these constraints, illustrating the risk of producing outputs that appear plausible but are in fact incorrect.

S3.5 Example of a prompting task

Task

Description: My little brother is 8 years old, is passionate about space, and does not understand the concept of a black hole. I would like to explain it to him, but I do not know how to adapt the explanation to his age.

Basic Prompt: *Explain what a black hole is.*

Example of a practical task used in Part 3 of the intervention, in which students practiced formulating and iteratively refining prompts.

S4. Automatic annotation and prompt validation

S4.1 Automatic annotation protocol

Both prompt evaluation and student answer evaluation relied on GPT-4o as an automatic rater of answer quality on an ordinal scale. In all cases, we followed the same validation protocol: before using the model to annotate a full dataset, we first evaluated it on a manually annotated subset and verified that its agreement with human raters met predefined reliability and non-inferiority criteria.

Because large language model outputs are stochastic, all evaluations of GPT-4o were performed using three independent runs with identical prompts and parameters, and all metrics were averaged across runs (Liu et al., 2025).

Two complementary evaluation frameworks were used:

(1) Inter-rater reliability. Agreement was quantified using Krippendorff’s α for ordinal data. To compare human-only agreement to agreement including GPT-4o, we computed the paired difference:

$$\Delta = \alpha_{\text{human+model}} - \alpha_{\text{human-only}}.$$

Confidence intervals for Δ were estimated using a non-parametric bootstrap procedure (2,000 resamples per run; 6,000 total). We adopted a non-inferiority framework with margin $\delta = -0.05$: the model was considered acceptable if the lower bound of the 90% confidence interval of Δ was greater than δ .

(2) Advantage-probability analysis. We used the method of Calderon et al. (2025), in which each human annotator is held out in turn and the model is compared against that annotator with respect to agreement with the remaining annotators. This yields a winning rate ω , defined as the proportion of annotators for whom the model performs at least as well as the held-out human.

An advantage probability is also computed, corresponding to the probability that the model agrees at least as well as a randomly chosen human annotator on a randomly chosen item. A tolerance parameter $\epsilon = 0.2$ was used to account for the lower cost of automatic annotation relative to human annotation, allowing the model to be judged equivalent even if it performed slightly worse.

S4.2 Validation of automatic annotation on prompt responses

To validate prompt effectiveness, we generated 10 responses per prompt using three LLMs available via DuckDuckAI: GPT-4o mini, Mistral Small 3, and Llama 3 70B. For each of the 12 exercises and each prompt type (valid vs. invalid), this resulted in a total of 720 generated responses.

Each generated response was rated on the same 3-level ordinal scale (0 = wrong or unusable, 1 = partially correct or incomplete, 2 = fully correct and sufficient), yielding per-prompt scores ranging from 0 to 20.

Using the protocol described in S2.1, we selected a stratified 10% subset of the prompt-response dataset (72 responses to ensure coverage of all exercises, prompt types, and models.) and had it annotated by three independent human annotators and by GPT-4o (three independent runs).

Inter-annotator agreement was higher when GPT-4o was included (Krippendorff's $\alpha = 0.825$) than for the human-only set ($\alpha = 0.763$), with $\Delta = +0.062$, 90% CI [+0.036, +0.092]. Advantage-probability analyses indicated that GPT-4o outperformed each human annotator ($\omega = 1.00$; advantage probability = 0.98).

These results confirmed that GPT-4o could be used as an automatic rater for the full prompt-response dataset.

S4.3 Discrimination score and task/model selection

To jointly evaluate valid prompts (target score = 20) and invalid prompts (target score = 0), we defined a discrimination score:

$$D = \frac{\text{Score}_{\text{valid}} + (20 - \text{Score}_{\text{invalid}})}{2}$$

Under this definition, $D = 20$ indicates perfect discrimination, $D = 10$ chance-level discrimination, and $D = 0$ the reverse of the intended pattern.

We used this annotation system to identify the model that best discriminates between valid and invalid prompts (i.e., a model capable of consistently producing responses that effectively address exercise questions with valid prompts, and conversely, responses that fail or struggle to do so with invalid prompts). We used this score to (i) select the model used in the classroom and (ii) identify exercises whose prompts required revision. Because GPT-4o mini and Mistral Small 3 showed similar discrimination scores, we selected GPT-4o mini, as it likely represents the model students most commonly interact with (via ChatGPT).

Some exercises initially showed suboptimal discrimination. We therefore revised problematic prompts and created two new exercises. The validation procedure for responses generated from these revised prompts and new tasks was repeated. Based on the resulting scores, exercises 5 and 9 were excluded.

The final exercise set achieved a mean discrimination score of 19.79. Table S3 reports exercise-level results.

Exercise	Score (valid)	Score (invalid)	Discrimination
2	20	0	20
3	20	0	20
6	20	0	20
4	20	0	20
10	20	0	20
8	20	0	20
13	20	0	20
14	20	0	20
12	20	0	20
11	20	0	20
1	19	0	19.5
7	19	3	18.0
9 †	20	8	16.0
5 †	20	10	15.0

Table S3. Exercise-level discrimination (GPT-4o mini)

S4.4 Cross-model robustness

During classroom sessions, some students had to switch from DuckDuckAI to ChatGPT or TalkAI due to rate limits, resulting in the use of GPT-4o, GPT-4o mini, GPT-4.1-nano, or GPT-3.5-turbo.

Using the same validation procedure (10 responses per prompt), we verified that the valid–invalid discrimination property was preserved across these models. Table S4 summarizes discrimination scores by model, with a mean discrimination score of 18.2 across models.

Model	Mean Overall	Valid-prompt mean	Invalid-prompt mean
GPT-4o-mini	19.25	19.50	1.00
GPT-4.1-nano-2025-04-14	18.08	18.67	2.50
GPT-3.5-turbo-0125	17.79	18.75	3.17

GPT-4o	17.71	17.17	1.75
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Table S4. Model-level discrimination summary

S4.5 Validation of automatic annotation on student answers

Using the same protocol as in S2.1, we evaluated GPT-4o on a random $\approx 10\%$ subset of student answers (70/696), annotated by four human raters and by GPT-4o (three independent runs). To obtain an individual performance score, each student’s response to the six exercises was rated on a three-point scale: 0 (incorrect), 1 (partially correct), and 2 (fully correct).

Inter-rater reliability for the human-only set was Krippendorff’s $\alpha = 0.740$, while quartets including GPT-4o yielded a mean $\alpha = 0.752$. The paired difference $\Delta = +0.012$ had a 90% confidence interval $[-0.026, +0.048]$, which lay entirely above the non-inferiority margin $\delta = -0.05$. The advantage-probability analysis further showed $\omega = 1.00$, with an advantage probability of approximately 0.89, indicating that GPT-4o matched or outperformed each individual human annotator. Together, these results confirmed that GPT-4o could be reliably used to score the full student answer dataset.

S5. Supplementary results

S5.1 Distribution of success rates

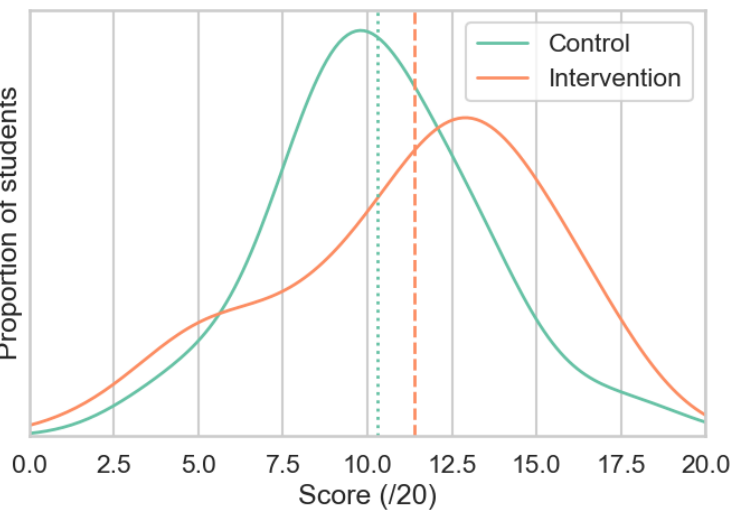


Figure S6. Distribution of overall performance scores (0–20) by group. Kernel density estimates show the distribution of per-student scores in the control (green) and intervention (orange) groups. Vertical dashed lines indicate group means.

S5.2 Distribution of individual sensitivity to prompt quality (d’)

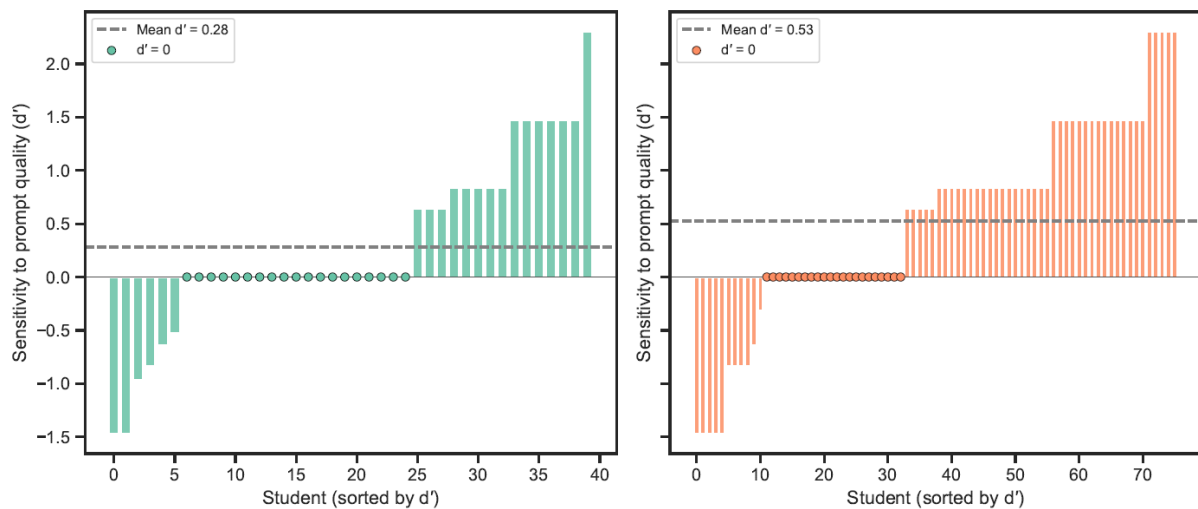


Figure S7. Distribution of individual sensitivity to prompt quality (d') in the control (left, blue) and intervention (right, orange) groups, with students sorted within each group. Each bar represents one student, and the dashed line indicates the group mean. The intervention group showed higher sensitivity overall (median = 0.73 vs. 0; mean = 0.53 vs. 0.28) and a larger proportion of students with positive sensitivity (56.6% vs. 37.5%). However, the difference between groups remained small and did not reach statistical significance (Mann–Whitney $p = .107$, Cliff’s $\delta = 0.18$; Fisher’s exact test on $d' > 0$: $p = .078$).

S5.3 Detailed analyses of prompt replacement

We examined how students accept/reject decisions influenced performance depending on prompt type and condition. For each student, we computed a within-subject contrast defined as performance after rejecting minus performance after accepting, separately for well-specified and underspecified prompts, and tested these contrasts against zero (using one-sided t-tests) and between groups (using Welch’s t-tests).

For underspecified prompts, this contrast was significantly positive in both groups (control: $t(14) = 2.74$, $p = 0.008$, $d = 0.71$; intervention: $t(40) = 2.34$, $p = 0.012$, $d = 0.37$), indicating that rejecting an underspecified prompt consistently improved score. The group difference was not significant (Welch’s $t(22.12) = 1.34$, $p = 0.194$).

For well-specified prompts, however, group patterns diverged. In the control group, rejecting well-specified prompts significantly reduced scores ($t(13) = -2.48$, $p = 0.014$, $d = 0.66$). In the intervention group, the same contrast tended to be positive, though it did not reach significance ($t(30) = 1.38$, $p = 0.089$). Importantly, the group difference was significant (Welch’s $t(25.32) = -2.83$, $p = 0.009$, $d = 0.91$), indicating opposite effects across groups. In other words, while untrained students were penalized for discarding useful prompts, trained students were often able to replace them without a performance cost.

S5.4 Detailed analyses of correctness judgments

We examined whether students were able to correctly evaluate the factual correctness of the AI's responses as a function of prompt type and experimental condition, using a trial-level logistic regression restricted to accepted prompts. Ideally, responses generated following well-specified prompts should be judged as correct, whereas responses generated following underspecified prompts should be judged as incorrect.

Descriptively, both groups showed higher accuracy for well-specified than for underspecified prompts, but this improvement was substantially larger in the intervention group. In the control group, accuracy increased from 50.0% to 67.4%, whereas in the intervention group it increased from 40.7% to 78.1%.

The model revealed a significant effect of prompt type in the control group ($\beta = 0.72$, $p = .001$, OR = 2.06, 95% CI [1.32, 3.22]), indicating that answers generated from well-specified prompts were more likely to be judged as correct than those generated from underspecified prompts. Crucially, this prompt-type effect was significantly larger in the intervention group, as indicated by a significant condition $\times$ prompt-type interaction ($\beta = 0.92$, $p = .012$, OR = 2.52, 95% CI [1.22, 5.19]). There was no significant main effect of condition for underspecified prompts ($\beta = -0.38$, $p = .27$).

S5.5 Detailed analyses of follow-up behavior

Finally, we analyzed how often students posed follow-up questions (at least one) to the LLM when the suggested prompt was accepted. For underspecified prompts, students in the intervention group were substantially more likely to ask follow-up questions than controls (59.2% vs. 27.9%). A Welch test on per-student proportions confirmed that this difference was highly significant ($t = 3.80$, $p = 0.0003$). The 95% confidence intervals around the group means did not overlap (control: [15.0%, 40.9%]; intervention: [48.7%, 69.6%]). For well-specified prompts, no group difference was observed. This indicates that the intervention specifically encouraged students to engage more critically with AI responses by prompting follow-up questions when outputs appeared inadequate.

S5.6 Additional Analyses on Attitudes and Metacognition

To test whether the Attitude–Fairness composite predicted performance and whether this relationship was moderated by metacognition, we fitted separate linear regression models in the control group and in the intervention group. In the control group, neither the composite score, nor metacognition, nor their interaction significantly predicted performance (all $p > .61$; model $R^2 = .015$). In the intervention group, the overall model was not significant ($R^2 = .049$), and neither the main effects of the composite score nor metacognition reached significance (both $p > .35$). The interaction between the composite score and metacognition showed only a marginal trend ($\beta = 0.19$, $p = .096$), indicating that the relationship between Attitude–Fairness and performance was not reliably moderated by metacognitive ability.

Overall, these results indicate that neither Attitude–Fairness scores nor their interaction with metacognition provide a robust explanation of individual differences in task performance, in either group.